

Your Turn: Design a Secure Workflow

Secure AI Workshop — Building a Secure Internal AI System for Higher Education

Group / Team: _____ Members: _____

How this works. In your group, apply the seven-layer architecture and the 70% confidence threshold to a real academic scenario. Pick a scenario, assess its risks, configure four key layers, and design the human checkpoint. Aim for *defensible, not perfect* — you will defend your design at the end.

ACTIVITY · STEP ZERO

Choose your scenario

Pick **one** as a group — you'll design and defend it.

☐ Scenario A

AI-Assisted Grading System

Generates feedback and suggested scores for student essays.

☐ Scenario B

IP-Protected Research Assistant

Summarizes unpublished manuscripts and identifies research gaps.

ACTIVITY · STEP 1 · RISK

Name two critical risks of using public AI 5 min

If your group ran this scenario on public ChatGPT instead of an internal system, what could go wrong? Give a concrete example for each.

Risk 1 — Data privacy & IP: *what could leak?*

Risk 2 — Integrity & accuracy: *where do hallucinations hurt?*

ACTIVITY · STEP 2 · CONFIGURE FOUR LAYERS

Apply the architecture 15 min

Be concrete — name real systems and real data sources.

L1 • Access *Who may use it, and what are their limits?*

L3 • Input Safety Which PII or keywords must be redacted?

L4 • RAG Which institutional sources will ground it?

L6 • Output Control How do you verify against the sources?

ACTIVITY • STEP 3

Design the human checkpoint 10 min

Note. The 70% threshold (and the 62% example) is an illustrative default, not a fixed standard — calibrate it per task on your own data. What matters is the principle: route uncertain outputs to a human.

THE 62% TEST

Your AI returns a result at 62% confidence.

It's below the 70% threshold. Map the exact steps the faculty reviewer takes before it reaches the end user.

GUARD THE REVIEWER

Now keep that reviewer from burning out.

Spell out your fatigue strategy — rotation, escalation, and what you deliberately choose not to review.

Reviewer's steps for the 62% result:

Fatigue-mitigation strategy:

ACTIVITY · EVALUATION

Evaluate your design

Total ____ / 20

Use these as a checklist while you design — not just at the end. Score each criterion out of 5.

CRITERION	PROFICIENT · 3–4	EXEMPLARY · 5	SCORE
Security & Privacy	Basic PII filters and user roles identified.	Full Layer-3 sanitization and local-first data sovereignty.	____ / 5
Grounding & Accuracy	Uses RAG, but sources are loosely defined.	Layer-4 RAG with citations and Layer-6 consistency checks.	____ / 5
HITL Effectiveness	A review process exists, but triggers are unclear.	Clear 70% threshold with a fatigue-mitigation strategy.	____ / 5
Institutional Value	Addresses a need, but ignores infrastructure cost.	Balances operational complexity against clear benefits.	____ / 5

PRESENT · DISCUSSION

Defend your design (2 minutes)

- Which scenario did you choose, and what was its single biggest risk?
- Which one layer did the most work to make it safe?
- Walk us through what happens to a 62% result.
- What did you deliberately choose NOT to review — and why is that safe?

Notes from discussion:

Optional hands-on lab. Want to run this design instead of just sketching it? Open the “Notebooks” folder (00–04) and run the secure pipeline yourself — PII redaction, RAG grounding, and the 70% threshold, all offline with no API key needed.

Bring AI inside the walls — controlled, grounded, and human-supervised.

A conceptual framework, not a finished product — the real test is the pilot. · Nguyen Thanh Tuan · nttuan@vku.udn.vn · Vietnam–Korea University of ICT, University of Danang